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ABSTRACT

We report extensions of a three-dimensional computational model for the liquid wall films formed in port-injected engines. The extensions incorporate effects associated with spray/wall interactions-including droplet splash, film spreading due to impingement forces, and motion due to film inertia. We also include a submodel for the effects of liquid expelled from valve seat areas when valves close. Implementation of the extensions in the KIVA-3V computer program is described, and results of KIVA calculations of open- and closed-valve injection in a realistic four-valve engine geometry are presented. Computed film locations agree qualitatively with those observed in laser-induced fluorescence measurements.

INTRODUCTION

This is a report of an extension to the computational model for liquid wall films that is presented in an earlier SAE paper [1]. This previous paper describes a wall film model that predicts film transport on complex surfaces, heating and vaporization of the film, and separation and re-entrainment of the liquid film at sharp corners. Film transport is determined by the action of shear stresses in the gas boundary layers above the films, by the tangential momentum deposition of sprays impinging on the films, and by the no-slip wall condition. Vaporization of the films is due to differences between the fuel vapor mass fraction in the bulk gas above the films and the vapor mass fraction at the surface of the film determined by the surface temperature of the film. The film surface temperature is, in turn, determined implicitly by a wall film temperature equation that accounts for heat transfer from the gas and the wall and heat addition due to an impinging spray. Mass, momentum, and energy transport in the gas boundary layers above the films are predicted by new wall functions for the turbulent boundary layers above vaporizing films. The wall film model in [1] is implemented in the three-dimensional hydrodynamics code KIVA-3V [2], using a computational particle method that has low numerical diffusion in its film transport calculation.

Reference [1] omitted a splash submodel, and we now rectify this omission. Splashing is important in port-fuel-injected engines, which are the main object of our study and where it is observed [3] that splash results in rebounding of a significant fraction of the spray from the impingement location and deposition of this fuel in films on neighboring valve and intake port wall surfaces. Splash is also an important physical phenomena in small-bore, direct-injected gasoline and diesel engines, as well as a variety of other industrial devices in which sprays impinge on solid surfaces.

When modeling the effects of splash on liquid fuel distributions in engines using a code such as KIVA, one requires detailed knowledge of the criterion for and results of splash. When a droplet impinges on a solid surface, which may be dry or wetted by a wall film, all its mass may be incorporated into the wall film or some of it may rebound from the surface in the form of new spray droplets. A detailed model for spray impingement, such as that in KIVA, requires not only the criterion for whether new spray droplets are formed, but the total mass associated with all rebounding droplets, as well as their sizes, velocities, and temperatures. In what follows the new spray droplets formed during splash will be called secondary droplets.

Only recently have detailed multidimensional models of impinging sprays been available, and perhaps because of the newness of these models little research has been done to provide the results of the splash process that are necessary in detailed models. Because of its complexity, direct hydrodynamics simulations of the splash process have been mostly qualitative in nature [4]. Most previous studies have been experimental and have concentrated on the criterion for whether splash occurs. A fairly recent review is contained in Ref. [5], which reports that there is little general agreement on splash criteria. More recently, however, two studies have been reported that begin to provide the information necessary for detailed models, and our splash model is based on the results of these two studies.

The first study, reported by Mundo, Sommerfeld and Tropea [6] and referred to subsequently as MST, is an experimental study of isolated liquid droplets impinging on dry solid surfaces. While varying impingement angles, liquid viscosities, surface tensions, and droplet sizes, the authors find a correlation between a single dimensionless splash parameter and the occurrence of splash. For fixed liquid properties, the splash parameter depends only on the droplet velocity normal to the surface and the droplet diameter. This same splash criterion has been proposed by others [7]. The authors also measure the sizes and velocities of the secondary droplets using phase Doppler anemometry.

In a second study, Yarin and Weiss (YW) [8] have reported a combined experimental and theoretical investigation of splashing droplets. The authors show that the threshold for splash corresponds to the formation of a kinematic discontinuity in the equations for a thin film. The velocity discontinuity, located at the boundary between fluid moving outward from the splash location and slower moving fluid on the wall surface, causes fluid to be ejected away from the surface in what is observed experimentally to resemble a crown. The crown is unstable and breaks up into splash product droplets. This scenario is corroborated experimentally by high speed photographs of the splash process. The experiments are performed with a train of droplets of varying frequency f impinging on a wet wall, and the authors also measure secondary droplet sizes and the total mass of secondary droplets. In contrast to the study of MST, YW find no correlation of splash threshold with incident droplet size and normal velocity for given liquid properties. Another parameter must enter in their experiment, and YW choose the train frequency f . The authors formulate and verify a splash criterion that, for fixed liquid properties, depends on the droplet velocity normal to the wall and the frequency f , but *not* on the droplet diameter.

In the next section of this report we give the MST and YW results in more detail and use them to formulate provisional splash criteria and secondary droplet property distributions that are necessary for the KIVA splash model. The splash criterion we propose depends on the film thickness, as well as the drop velocity normal to the wall, the drop diameter, and the liquid density, viscosity, and surface tension coefficient. Our criterion agrees with that of MST for dry surfaces. We argue that the experimental result of YW is of little use in a general splash model because of the special experimental arrangement. In formulating the dependence of the splash criterion on liquid film thickness, however, we do use the argument of YW that the splash criterion should depend on a splash "Mach number," based on the capillary wave speed, exceeding a critical value. After describing the splash model we tell how it is implemented in the KIVA code.

In addition to incorporating the splash model into KIVA, we also have found it necessary to include two terms in the wall film momentum equation that were ignored in our

previous work [1]. The first of these is the wall film inertia term, which can be important near impingement sites [9]. The second term arises because of wall film pressure gradients resulting from the normal momentum of the impinging spray. These pressure gradients cause spreading of wall film away from impingement sites. The revised wall film momentum equation is given in the next section, and its numerical solution method in KIVA is described.

An additional phenomenon of importance in port-fuel-injected engines is the expelling of liquid from valve seat areas when valves close. This liquid then breaks up into spray droplets in either the intake port or the cylinder. In the next section we describe a submodel for predicting the amounts and velocities of liquid ejected into the intake port and cylinder when valves close.

Finally, we give the results of a group of calculations simulating experiments performed by Witze [3,10] in a four valve port-fuel-injected engine.

EXPERIMENTAL RESULTS AND THE SPLASH MODEL

THE SPLASH CRITERION – Mundo, et al. [MST, 6] measured droplets impinging on dry smooth and rough surfaces, varying fuel properties, drop sizes, and impingement angles and velocities. They found that for splash to occur on smooth surfaces a certain dimensionless parameter K had to be greater than a critical value of 57.7:

$$K = Oh \cdot Re^{1.25} > 57.7 \quad (1)$$

where Oh is the Ohnesorge number,

$$Oh = \frac{\mu}{\sqrt{\rho\sigma D}}, \quad (2)$$

Re is the Reynolds number based on the drop's velocity w_0 normal to the impact surface,

$$Re = \frac{\rho w_0 D}{\mu}, \quad (3)$$

σ is the surface tension coefficient, μ is the liquid viscosity, ρ is the liquid density, and D the incident drop's diameter. We note that by definition of the Weber number We , we can also express

$$K = We^{0.5} \cdot Re^{0.25}. \quad (4)$$

MST also measured the size and velocity distributions of secondary droplets. The sizes of secondary droplets were dependent on the value of K , with smaller droplets and narrower distributions being produced as K was increased. The velocities of secondary droplets scaled with incident drop velocities and were independent of K for fixed incident velocity. Mean secondary droplet normal and tangential velocities scaled with incident drop normal

and tangential velocities, respectively, while the widths of both the secondary droplet normal and tangential velocity distributions scaled approximately with incident drop normal velocity.

Yarin and Weiss [YW,8] performed experiments in which a monodisperse train of droplets impacted a solid surface wetted by the impacts of previous droplets in the train. They measured the conditions under which splash occurred, secondary droplet sizes, and the total mass of secondary droplets. YW found that splash occurred if and only if a dimensionless impact velocity u exceeded 17:

$$u = \frac{w_0}{\left(\frac{\sigma}{\rho}\right)^{1/4} \left(\frac{\mu}{\rho}\right)^{1/8} f^{3/8}} \gtrsim 17, \quad (5)$$

where f is the frequency with which the incident droplets impact the surface. Note that the incident droplet diameter does not enter into this correlation. We believe that the correlation (5) cannot be used for practical spray situations because of the special experimental conditions of YW's experiments. The splash event was influenced by a pre-existing wall film whose thickness h and state of motion were both functions of frequency f . Of course, both of these influences are present in a real spray impinging on a surface, but the axisymmetric impact geometry of YW's experiment is not.

YW do give theoretical justification for their correlation, which we now briefly describe and which gives some insight into the splash event. For axisymmetric film flow in which only surface tension forces are considered, the film momentum equation is

$$\rho h \left(\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial r} \right) = \sigma \frac{\partial}{\partial r} \left(h \frac{\partial^2 h}{\partial r^2} \right), \quad (6)$$

where v is the film velocity tangent to the wall and h is the film thickness. If one nondimensionalizes h by the initial film thickness h_0 , v by the impact velocity w_0 , time t by $1/f$, and r by w_0/f , as is done by YW, then the film momentum equation becomes

$$h \left(\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial r} \right) = \frac{1}{R} \frac{\partial}{\partial r} \left(h \frac{\partial^2 h}{\partial r^2} \right), \quad (7)$$

where all quantities are dimensionless and

$$R = \frac{\rho}{\sigma h_0} \frac{w_0^4}{f^2}. \quad (8)$$

The quantity R is the ratio of liquid inertial forces to surface tension forces that tend to smooth the liquid surface, and YW argue that splash occurs if the value of R exceeds a critical value. The YW splash criterion (5) is obtained if one substitutes for the film thickness in (8) the dimensionally correct expression $h_0 = [\mu/(\rho f)]^{0.5}$.

In this work we are interested in extending the isolated-drop dry-wall correlation of MST to the isolated-drop wet-wall case, and a modification of the YW argument provides a reasonable basis for doing this. If an isolated drop impacts a wet surface, then we argue the relevant time and length scales for the splash process should be D/w_0 and D , respectively. Nondimensionalizing the film momentum equation using these new time and length scales results in

$$h \left(\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial r} \right) = \frac{1}{E^2} \frac{\partial}{\partial r} \left(h \frac{\partial^2 h}{\partial r^2} \right) \quad (9)$$

where E is a splash Mach number based on the impact velocity and the capillary wave speed of a wave of length D :

$$E = \frac{w_0}{\frac{1}{D} \sqrt{\frac{\sigma h_0}{\rho}}}. \quad (10)$$

We would like to use the splash criterion $E > E_{crit}$, but this gives meaningless results as the film thickness goes to zero. If when h_0 goes to zero, however, in place of h_0 we use a boundary layer thickness

$$\delta_{bl} = \frac{D}{Re^{1/2}} \quad (11)$$

then the dry wall criterion of MST is recovered with the quantity $E = K$. Our splash criterion is obtained by using the sum of the film thickness and the boundary layer thickness, in place of h_0 , in the inequality $E > E_{crit}$:

$$E^2 = \frac{\rho w_0^2 D}{\sigma} \cdot \frac{1}{\min\left(\frac{h_0}{D}, 1\right) + \frac{\delta_{bl}}{D}} > (57.7)^2. \quad (12)$$

We also assume that further increases of film thickness h_0 above the value D no longer influence the splash event. The criterion (12) reflects the following physically plausible trends: the presence of a film on the wall suppresses splash by viscous damping of impact energy and by radiating energy away from the impact site through capillary waves.

SECONDARY DROPLET PROPERTIES – In addition to a splash criterion, we need expressions for the total mass, the size distribution, and the velocity distribution of secondary droplets. These are chosen to conform to the experimental results of MST and YW, an energy consistency result, and the splash direction results of [11]. We note there is still considerable uncertainty in these expressions even after they are constrained by the experimental and energy consistency results, and there is a critical need for further experimental measurements of splash parameters.

Experimental results for total mass of secondary droplets are only reported by YW. YW report that, above the

critical impact velocity u of 17, the ratio of total mass M of secondary droplets to incident drop mass M_0 rises from zero to a value of about 0.75 for u approximately equal 21 and thereafter remains about constant. It is difficult to make a correspondence between YW's dimensionless impact parameter u and our E , for reasons discussed above. The experiment does seem to indicate that M/M_0 is greater than 0.0 and less than 0.75 for only a narrow range of u (or E). Based on this data we use the following expression for M/M_0 :

$$\frac{M}{M_0} = \begin{cases} 1.8 \times 10^{-4} (E^2 - E_{crit}^2) & \text{for } E_{crit}^2 < E^2 < 7500 \\ 0.75 & \text{for } 7500 < E^2 \end{cases} \quad (13)$$

Size distributions of secondary droplets are reported both by MST and by YW, who give probability density functions (pdf's) of secondary droplet radius r divided by incident drop radius r_0 . Their results are consistent with a Nukiyama-Tanasawa [12] size distribution:

$$f(r) = \frac{4}{\sqrt{\pi}} \frac{r^2}{r_{max}^3} \exp \left[- \left(\frac{r}{r_{max}} \right)^2 \right], \quad (14)$$

where r_{max} is the droplet radius at which the pdf achieves its maximum value. The value of r_{max}/r_0 is a function of E that we approximate by

$$\frac{r_{max}}{r_0} = \max \left(\frac{E_{crit}^2}{E^2}, \frac{6.4}{We}, 0.06 \right). \quad (15)$$

MST only report size distributions for E close to E_{crit} and their results are approximated by the first argument in the max of Eq. (15). The second argument limits the sizes of secondary droplets at high impact Weber numbers We and arises from an energy consistency requirement given below. In both experimental studies r_{max}/r_0 never is less than 0.06, which gives rise to the third argument in (15).

Secondary droplet velocity distributions are reported by MST. Mean secondary droplet normal and tangential velocities scaled with incident drop normal and tangential velocities, respectively, while the widths of both the secondary droplet normal and tangential velocity distributions scaled approximately with incident drop normal velocity. We approximate the observed distributions by choosing secondary droplet velocities w according to

$$\mathbf{w} = w' \mathbf{n} + (0.12w_0 + v')(\cos\psi \mathbf{e}_t + \sin\psi \mathbf{e}_p) + 0.8v_0 \mathbf{e}_t \quad (16)$$

where $\mathbf{n}$ is the unit normal to the wall surface; $\mathbf{e}_t$ is the unit vector tangent to the surface and in the plane of $\mathbf{n}$ and the incident drop velocity; $\mathbf{e}_p = \mathbf{n} \times \mathbf{e}_i$; v_0 is the component of the incident drop velocity tangent to the surface; and w' , v' , and ψ are random variables chosen

as follows. The quantity w' is the normal component of the secondary droplet velocity and is chosen from the Nukiyama-Tanasawa distribution:

$$P(w') = \frac{4}{\sqrt{\pi}} \frac{(w')^2}{w_{max}^3} \exp \left[- \left(\frac{w'}{w_{max}} \right)^2 \right], \quad (17)$$

where $w_{max} = 0.2w_0$ and w_0 is the incident drop normal velocity. The quantity v' is the fluctuating component of the secondary droplet tangential velocity, chosen from the Gaussian distribution:

$$G(v') = \frac{1}{\sqrt{2\pi}\delta^2} \exp \left[- \frac{(v')^2}{2\delta^2} \right], \quad (18)$$

where $\delta = 0.1w_0$. Finally, ψ is the angle that the fluctuating tangential velocity makes with the vector $\mathbf{e}_t$ in the plane of the wall. ψ lies in the interval $(-\pi, \pi)$ and is chosen according to the distribution suggested in [11]:

$$H(\psi) = \begin{cases} \frac{1}{2\pi} & \beta = 0 \\ \frac{\beta/\pi}{2(e^\beta - 1)} \exp \left[\beta \left(1 - \frac{|\psi|}{\pi} \right) \right] & \beta > 0 \end{cases}, \quad (19)$$

where β is a parameter related to the impact angle α by

$$\sin\alpha = \frac{e^\beta + 1}{e^\beta - 1} \frac{\beta^2}{\beta^2 + \pi^2}$$

and α is the angle between the incident drop velocity and the wall normal $\mathbf{n}$.

ENERGY CONSISTENCY REQUIREMENT

A lower bound on the sizes of the secondary droplets produced by the splash event can be derived by using the requirement that the total energy of the secondary droplets, including their surface energy, cannot exceed the incident drop energy. We now indicate how this can be done. An extension and refinement of the argument of this section is given in Ref. [13].

The total energy of the incident drop E_0 is given by

$$E_0 = \frac{1}{2} M_0 (w_0^2 + v_0^2) + 4\pi r_0^2 \sigma. \quad (20)$$

Assuming the random variables r , w' , v' , and ψ are statistically independent, the total energy of the secondary droplets E is

$$E = N \iiint \left\{ 4\pi r^2 \sigma + \frac{2}{3} \pi r^3 \rho \left[(w')^2 + (0.8v_0 + (0.12w_0 + v') \cos \psi)^2 + ((0.12w_0 + v') \sin \psi)^2 \right] \right\} f(r) P(w') G(v') H(\psi) dr dw' dv' d\psi, \quad (21)$$

where N is the total number of secondary droplets. N is determined from the requirement that

$$N \int f(r) \frac{4}{3} \pi r^3 \rho dr = M. \quad (22)$$

Using Eq. (14), requirement (22) can be seen to give

$$N = \frac{M}{\frac{16}{3} \pi^{1/2} \rho r_{\max}^3}. \quad (23)$$

For energy consistency it is required that $E < E_0$.

An expression for E is obtained by a straightforward, but lengthy, derivation involving taking integrals of the distributions (14), (17), (18), and (19). One obtains

$$E = \frac{1}{2} M \left[\frac{9}{4} \pi^{1/2} \frac{\sigma}{\rho r_{\max}} + 0.0844 w_0^2 + 0.64 v_0^2 + 0.192 v_0 w_0 \sin \alpha \right]. \quad (24)$$

After assuming that $M = 0.75 M_0$ (which is usually the case), substituting the expressions for E and E_0 in Eqs. (24) and (20) into the requirement that $E < E_0$, and some algebraic manipulation, one obtains the following constraint on $r_{\max}$:

$$\begin{aligned} & 7.98 \frac{\sigma}{\rho w_0^2 D} \frac{r_0}{r_{\max}} + 0.192 \tan \alpha \sin \alpha \\ & < 1.25 + 0.69 \tan^2 \alpha + 16 \frac{\sigma}{\rho w_0^2 D} \end{aligned} \quad (25)$$

where α is the impact angle defined above. One can easily show it is sufficient to have

$$\frac{r_{\max}}{r_0} > \frac{6.4}{We}$$

in order for Eq. (25) to be satisfied, which is our energy consistency condition.

MODIFICATIONS TO THE FILM MOMENTUM EQUATION FOR SPRAY/WALL INTERACTION

In Ref. [1] two assumptions were made that are valid outside of the spray/wall interaction region but that are violated when sprays impinge on wall films. These are that the film inertial terms and film pressure gradient term are negligible in the wall film momentum equation. In this

section we give the revised film momentum equation that includes these terms. We continue to assume that the velocity profile within the film varies linearly within the film, with mean value $\mathbf{u}_f$ that satisfies $(\mathbf{u}_f - \mathbf{u}_w) \cdot \mathbf{n} = 0$, where $\mathbf{u}_w$ is the wall velocity. We also include the gravitational term because of its importance when films are allowed to reside on walls for the relatively long periods of time between injector events.

The revised wall film momentum equation is

$$\begin{aligned} \rho h \left\{ \frac{\partial \mathbf{u}_f}{\partial t} + [(\mathbf{u}_f - \mathbf{u}_w) \cdot \nabla_s] \mathbf{u}_f \right\} + h \nabla_s p_f = \tau_w \mathbf{t} \\ - \mu_f (T_f) \frac{\mathbf{u}_f - \mathbf{u}_w}{h/2} + \dot{\mathbf{P}}_{imp} - (\dot{\mathbf{P}}_{imp} \cdot \mathbf{n}) \mathbf{n} + \\ \dot{M}_{imp} [(\mathbf{u}_w \cdot \mathbf{n}) \mathbf{n} - \mathbf{u}_f] + \delta p_f \mathbf{n} + \rho h \mathbf{g} \end{aligned} \quad (26)$$

where

$\frac{\partial}{\partial t}$	= time-rate-of-change in the wall frame of reference,
∇_s	= surface gradient operator,
τ_w	= shear stress on the gas-side of the wall film,
$\mathbf{t}$	= unit tangent to the wall in the direction $\mathbf{u}_f - \mathbf{u}_w$,
μ_f	= film viscosity (temperature-dependent),
T_f	= mean film temperature,
$\dot{\mathbf{P}}_{imp}$	= momentum source/area due to impingement,
$\dot{M}_{imp}$	= liquid mass source/area due to impingement,
p_f	= film (impingement) pressure, and
δp_f	= pressure difference across film to bring about $(\mathbf{u}_f - \mathbf{u}_w) \cdot \mathbf{n} = 0$, and
$\mathbf{g}$	= acceleration due to gravity.

The revised mass and momentum sources due to impingement are given by

$$\dot{M}_{imp} = - \iiint \frac{4}{3} \pi r^3 \rho \mathbf{v} \cdot \mathbf{n} f(\mathbf{x}_s, \mathbf{v}, r, T_d, t) d\mathbf{v} dr dT_d \quad (27)$$

and

$$\dot{\mathbf{P}}_{imp} = - \iiint \frac{4}{3} \pi r^3 \rho \mathbf{v} \mathbf{v} \cdot \mathbf{n} f(\mathbf{x}_s, \mathbf{v}, r, T_d, t) d\mathbf{v} dr dT_d, \quad (28)$$

where f is the droplet distribution function [14] at a point $\mathbf{x}_s$ on the wall.

In Eq. (26) the new terms are on the left-hand side of the equation: the inertial terms and the film pressure gradient term. The expression in braces is the material time-

derivative of the film velocity, the time-rate-of-change of the film velocity following a film mass element. The mean film pressure p_f is assumed to arise entirely from the impingement of the spray:

$$p_f = \dot{M}_{imp} \mathbf{u}_w \cdot \mathbf{n} - \dot{\mathbf{P}}_{imp} \cdot \mathbf{n}. \quad (29)$$

Thus we are ignoring contributions to p_f due to the gas-side pressure and due to surface tension. The latter are responsible for the capillary waves included in the analysis of YW [8] but contribute negligibly to changing the mean film velocity.

MODIFICATIONS TO THE NUMERICAL SOLUTION PROCEDURE

The modifications to the numerical method for wall films occur principally in the difference approximation to the film momentum equation. In our first publication of this model [1], the film inertial terms were not included in the film momentum equation, and as a result we could numerically solve for one film velocity for each wall face. Film particles on the same wall face then moved with the same velocity. When the inertial terms are included, this approximation cannot be used because each film particle must remember its past velocity. Hence we now solve the momentum equation for each film particle. The difference approximation to the momentum equation (26) for wall film particle p is

$$\begin{aligned} \rho h_\alpha \frac{\mathbf{u}_p^{n+1} - \mathbf{u}_p^n}{\Delta t} + h_\alpha (\nabla_s p_f)_\alpha &= \tau_{w,\alpha} \mathbf{t}_\alpha - \mu_\ell \frac{\mathbf{u}_p^{n+1} - \mathbf{u}_w}{h_\alpha / 2} \\ &+ \dot{\mathbf{P}}_{imp,\alpha} - \dot{M}_{imp,\alpha} \mathbf{u}_p^{n+1} \\ &+ F_{n,\alpha} \mathbf{n}_\alpha + \rho h_\alpha \mathbf{g} \end{aligned} \quad (30)$$

where α is the wall face on which particle p resides. The quantities h_α , $\tau_{w,\alpha}$, $\mathbf{t}_\alpha$, and $\mathbf{n}_\alpha$ are, respectively, the film height, the gas shear stress on the top of the film, the unit vector in the direction of gas motion on top of the film, and the unit normal to the wall surface on face α . All normal forces on the film have been accumulated in the quantity $F_{n,\alpha}$. We defer until later the approximations to the film pressure p_f , the gradient of p_f on face α , and revised approximations to the mass and momentum impingement sources $\dot{M}_{imp,\alpha}$ and $\dot{\mathbf{P}}_{imp,\alpha}$. $\tau_{w,\alpha}$ is approximated as in Ref. [1].

The quantity $F_{n,\alpha}$ is determined by the constraint

$$(\mathbf{u}_p^{n+1} - \mathbf{u}_w) \cdot \mathbf{n}_\alpha = 0. \quad (31)$$

By using Eqs. (30) and (31) to solve for $F_{n,\alpha}$ and substituting the resulting expression into Eq. (30) one can solve explicitly for the advanced-time film particle velocity:

$$\begin{aligned} \mathbf{u}_p^{n+1} &= \left\{ \frac{\rho h_\alpha}{\Delta t} \mathbf{u}_p^n + \tau_{w,\alpha} \mathbf{t}_\alpha + \frac{2\mu_\ell}{h_\alpha} \mathbf{u}_{wall} + \mathbf{S}_\alpha \right. \\ &\quad - \left[\frac{\rho h_\alpha}{\Delta t} (\mathbf{u}_p^n - \mathbf{u}_{wall}) \cdot \mathbf{n}_\alpha + \mathbf{S}_\alpha \cdot \mathbf{n}_\alpha \right. \\ &\quad \left. \left. - \dot{M}_{imp,\alpha} \mathbf{u}_{wall} \cdot \mathbf{n}_\alpha \right] \mathbf{n}_\alpha \right\} \\ &\quad / \left[\frac{\rho h_\alpha}{\Delta t} + \frac{2\mu_\ell}{h_\alpha} + \dot{M}_{imp,\alpha} \right], \end{aligned} \quad (32)$$

where $\mathbf{S}_\alpha$ is given by

$$\mathbf{S}_\alpha = \dot{\mathbf{P}}_{imp,\alpha} - h_\alpha (\nabla_s p_f)_\alpha + \rho h_\alpha \mathbf{g}. \quad (33)$$

One can verify that if inertial, film pressure gradient, and gravitational terms are negligible then Eq. (32) reduces to the approximation given in Ref. [1].

The film pressure gradient on face α is approximated as follows. First, the film pressure on each face α is calculated by

$$p_{f,\alpha} = \dot{M}_{imp,\alpha} \mathbf{u}_w \cdot \mathbf{n}_\alpha - \dot{\mathbf{P}}_{imp,\alpha} \cdot \mathbf{n}_\alpha. \quad (34)$$

Then the film pressure gradient is approximated by

$$(\nabla_s p_f)_\alpha = \frac{1}{|\mathbf{A}_\alpha|} \sum_{k=1}^4 (p_f)_k \mathbf{t}_k |\mathbf{x}_{k+1} - \mathbf{x}_k|, \quad (35)$$

where the sum is over the four edges k of face α , $\mathbf{A}_\alpha$ is the area projection vector of face α , $(p_f)_k$ is the average of the film pressures on the two faces on either side of edge k , $\mathbf{x}_k$ and $\mathbf{x}_{k+1}$ are the locations of the two vertices at the ends of edge k , and $\mathbf{t}_k$ is the unit tangent to the surface in the direction normal to edge k . $\mathbf{t}_k$ is approximated by

$$\mathbf{t}_k = \frac{(\mathbf{x}_{k+1} - \mathbf{x}_k) \times \mathbf{A}_k}{|(\mathbf{x}_{k+1} - \mathbf{x}_k) \times \mathbf{A}_k|} \quad (36)$$

where $\mathbf{A}_k$ is the average of the area projection vectors of the two faces on either side of edge k . Equation (35) is an approximation to the following definition of the surface gradient operator:

$$\nabla_s q = \lim_{A \rightarrow 0} \frac{1}{A} \oint_C q d\mathbf{l} \times \mathbf{n} \quad (37)$$

where C is a closed contour on the surface enclosing area A , and q is a scalar surface variable.

The revised forms of the mass and momentum impingement sources $\dot{M}_{imp,\alpha}$ and $\dot{\mathbf{P}}_{imp,\alpha}$ are now given. The mass impingement source now includes only contributions from spray particles whose droplets stick to the wall face α this timestep

$$\dot{M}_{imp,\alpha} = \left(\sum_{\substack{p \text{ stick} \\ \text{on face } \alpha \\ \text{this } \Delta t}} \rho V_p \right) / (|\mathbf{A}_\alpha| \Delta t), \quad (38)$$

where V_p is the liquid volume associated with particle p . The momentum impingement source, on the other hand, includes contributions from both sticking and splashing spray particles that impinge on face α this timestep

$$\dot{\mathbf{P}}_{imp,\alpha} = \left(\sum_{\substack{p \text{ impinge} \\ \text{on face } \alpha \\ \text{this } \Delta t}} \rho V_p (\mathbf{u}_p^{n+1} - \mathbf{u}_p^n) \right) / (|\mathbf{A}_\alpha| \Delta t). \quad (39)$$

Consistent with the remainder of the stochastic spray model, if a spray particle splashes, its associated droplets are assigned one size r_p^{n+1} and velocity $\mathbf{u}_p^{n+1}$ chosen randomly from the size and velocity distributions given for splash product droplets in the section, "Secondary Droplet Properties."

VALVE-SEAT LIQUID-EXPELLING SUBMODEL

In this section we document a submodel for the expelling of liquid from valve seat areas when valves close. We give a brief derivation of the submodel equations, which assume viscous flow between two parallel plates with pressure boundary conditions imposed at the ends of the plates. The equations give liquid mass flow rates into the intake port and cylinder that depend upon the pressure difference between the port and cylinder and upon the rate at which the valve lift is changing. Then the numerical implementation of the valve-seat expelling model in KIVA-3V is briefly described.

When the valve lift has decreased to a value so that liquid completely fills the gap between the valve and valve seat, the liquid-expelling submodel is activated. We assume the distance $h(t)$ between the valve and seat is known, ignore gradients and flow in the direction parallel to the valve rim, and assume the flow can be approximated as planar and viscously dominated. We must obtain equations for the average velocities u_1 and u_2 at either end of the planar channel formed by the valve and seat.

One equation is obtained simply from mass conservation; that is, the rate of volume decrease between the valve and seat must equal the rate at which liquid volume is fluxed out of the channel:

$$L \frac{dh}{dt} = -h(u_2 - u_1) \quad (40)$$

where L is the channel length. Note that the streamwise coordinate x increases from end 1 to end 2, and thus if u_2 is positive, the flow is out of the channel, whereas if u_1 is positive, flow is into the channel.

A second equation is obtained from the liquid momentum equation. Since the flow is dominated by viscosity, a parabolic velocity profile pertains at each streamwise location. The average velocity $u(x)$ at each streamwise location is proportional to the local pressure gradient:

$$u(x) = -\frac{h^2}{12\mu} \frac{\partial p}{\partial x}. \quad (41)$$

By assuming the average velocity varies linearly between the velocities u_1 and u_2 at either end of the planar channel and integrating Eq. (41), we obtain

$$\frac{p_2 - p_1}{L} = -12 \frac{\mu}{h^2} \frac{u_1 + u_2}{2} \quad (42)$$

where p_1 and p_2 are the pressures at either end of the channel. Simultaneous solution of Eqs. (40) and (42) for velocities u_1 and u_2 then gives

$$u_1 = \frac{L}{2h} \frac{dh}{dt} - \frac{p_2 - p_1}{L} \frac{h^2}{12\mu} \quad (43a)$$

and

$$u_2 = -\frac{L}{2h} \frac{dh}{dt} - \frac{p_2 - p_1}{L} \frac{h^2}{12\mu} \quad (43b)$$

The implementation of the valve-seat expelling submodel is now described. The submodel is activated on a valve face α when the perpendicular distance between the valve and seat becomes less than the sum of the film heights h_α on the valve rim and seat. When the submodel is activated, the computational film particles on the valve face are changed to "expel" particles that no longer communicate with the gas. Each expel particle is initially assigned a liquid volume V_α equal to the average volume associated with all the particles on that wall face. As the valve closes, for each computational timestep we inject a number of expel particles N_{inj} equal to

$$N_{inj} = \text{Int} \left[\frac{(h_\alpha - h) A_\alpha}{V_\alpha} \right] \quad (44)$$

where A_α is the valve face area and Int is the largest integer function. The quantity h_α is then updated by

$$h_\alpha = h_\alpha - \frac{N_{inj} V_\alpha}{A_\alpha}. \quad (45)$$

For each of the N_{inj} particles, a random choice is made to determine whether it is injected into the intake port or cylinder, based on the velocities calculated by Eqs. (43a)

and (43b). The droplets associated with the resulting spray particles are given initial velocities of magnitude u_1 or u_2 , depending on which side they have been injected, and initial diameters equal to the current value of $h(t)$. Breakup of the resulting droplets into smaller ones is then calculated by the TAB breakup model [14].

EXAMPLE CALCULATIONS

In this section we discuss the results of two KIVA-3V calculations of port fuel injection (PFI) in a contemporary dual overhead camshaft (DOHC) 4-valve gasoline engine with canted valves, a pentroof combustion chamber, and a low-pressure dual spray injector. The discussion focuses on the differences observed between closed-valve injection (CVI) and open-valve injection (OVI) fueling schedules for a simulated cold start. These results are also compared with experimental laser-induced fluorescence (LIF) video images [3], [10]. The latest KIVA-3V version [15] was used, which incorporates all the features of the wall film model described above.

ENGINE AND COMPUTATIONAL PARAMETERS – The engine parameters are listed in Table 1. Note that all references to crank angle degrees (CAD) are relative to a value of 0 at top dead center (TDC) of the intake stroke, except where noted in figure captions for experimental results.

The engine was run at 1,300 rpm in motored (nonfiring) mode at room temperature (20 *degr* C), 40 kPa MAP, using isooctane (2,2,4-trimethylpentane) as the fuel. The fuel rate was 0.036 ml (24.9 mg)/injection in a square-wave pulse with a constant 2,100 cm/s injection velocity.

In the experiment, a window in the piston crown provided optical access to 80% of the combustion chamber. A dopant was added to the isooctane to enhance the fluorescence induced by a Nd-YAG laser that accessed the window through a set of lenses and mirrors. The LIF images were then captured by an intensified video camera with the piston at BDC of the intake stroke. It has been determined that fluorescence from vapor phase fuel is much weaker than that from liquid fuel. Because of this, vapor is not detected using this technique, which is convenient for the study of fuel films. However, the films are sparse, and thus the LIF signals are weak, and it was necessary to run the engine through 20 cycles before the films were sufficiently thick to record. Because the imaging technique is a line-of-sight method, it is not possible to explicitly distinguish between films located on the head and those on the window in the piston crown.

Table I. DOHC 4-Valve PFI Engine Parameters

Compression Ratio	9.5:1
Bore	9.2 cm
Stroke	8.5 cm
Connecting Rod	14.7 cm
RPM	1,300
Wall and Valve Temperature	20C
Fuel	Isooctane (2,2,4-Trimethylpentane)
Initial Fuel Temperature	20C
Inlet Air Temperature	20C
MAP	40 kPa
Valve Timings (after TDC of the intake stroke):	
IVO	-12 CAD
EVC	1 CAD
IVC	217 CAD
EVO	492 CAD
Injection Parameters	
Open-Valve Injection	
SOI	26 CAD
EOI	120 CAD
Closed-Valve Injection	
SOI	166 CAD
EOI	260 CAD
Injected Mass	24.9 mg
Injection Velocity	2,100 cm/s
Cone Angles	12°
Angle Between Dual Sprays	25°
Sauter Mean Diameter	140 μm

The engine timings used in this study are shown in Fig. 1. For true open-valve injection there is a constraint to the injection duration. For this reason, we injected for 12 ms or 94 CAD, with start-of-injection (SOI) at 26 CAD for the OVI case. This is conservatively short but should not compromise having pure OVI. The same 94 CAD duration was used for the CVI case, with SOI at 166 CAD. Theoretically, the fuel would arrive at the intake valves 24 CAD later, while the valves are still open. However, at 190 CAD the valves are closing, and the intake flow has reversed. In the KIVA-3V calculation, nearly 32 CAD are required for the fuel to actually reach the valves. This SOI was chosen to allow the maximum amount of time for fuel vaporization in the intake port before the intake valves reopen on the next engine cycle.

A perspective view of the KIVA-3V computational mesh is shown in Fig. 2. To take advantage of a plane of symmetry in this engine, only one half of the full geometry is zoned. As a result, the mesh includes only one intake and one exhaust valve and requires only 53% as many vertices as a full circle mesh. Preliminary calculations using a full circle mesh and a half circle mesh gave results that were virtually identical to one another. The computational parameters are summarized in Table II.

Table II. KIVA-3V Computational Parameters

Maximum Number of Active Vertices	26,673
Number of Computational Particles	1,000
Starting CA	-15 CAD
Problem Duration	
-15° to 705° (Open-Valve Injection)	
-15° to 1620° (Closed-Valve Injection)	
Number of Time Steps	
4,645 (Open-Valve Injection)	
10,568 (Closed-Valve Injection)	
HP-9000 Computer Time	
Open-Valve Injection	22.89 hours
Closed-Valve Injection	48.31 hours

CLOSED-VALVE INJECTION RESULTS – We now discuss the computational results, first for the CVI case. The KIVA-3V graphics in Fig. 3 show the location and thickness of fuel films on the bottom of the head and valves and on the piston crown at two different crank angles. For comparison with the experimental LIF images [10] in Fig. 4, the views include a mirror image of the unzoned half of the geometry and are oriented so that the intake valves are on the right. Because the films are sparse, the LIF signals are very weak. The experiment required perhaps 20 engine cycles to see distinguishable films, which is unacceptable from a computation time standpoint. Nevertheless, several general experimental and computational observations about CVI are that films form on the edge of the intake valve faces, that there is no appreciable film on the exhaust side of the cylinder, and that large droplets impinge on the piston crown.

Additional useful information is gained and inferences from Figs. 3 and 4 are confirmed from graphs in Figs. 5-8 of the calculated histories of the global mass of fuel liquid and vapor through the two engine cycles. The graph of fuel history in the intake port (Fig. 5) confirms that most of the fuel injected becomes film on the intake valve tops and stems, with a much smaller mass becoming film on the port walls.

The graph of vapor history (Fig. 6) indicates that approximately 2 mg of vapor are formed in the intake port each engine cycle. This then enters the cylinder and is augmented by enhanced film evaporation during the

intake process. However, most of the vapor is subsequently lost out the exhaust port, and one can surmise that many engine cycles would be required before the vapor in the cylinder at the time of ignition becomes rich enough to burn. Indeed, an examination of the masses of fuel vapor and O_2 in the cylinder indicates that the overall equivalence ratio would be 1.08 at an approximate ignition time (1052 CAD) if all the fuel had entered the cylinder and vaporized, but the actual average equivalence ratio is only 0.15. A smaller amount of fuel vapor (0.6-0.8 mg) is lost out the intake port but is reintroduced when the flow reverses in the next intake stroke.

Of the 0.63 mg of spray created in the cylinder from film expelled from the intake valve seats at IVC (Fig. 7), about 0.5 mg vaporizes between IVC (937 CAD) and EVO (1212 CAD), as shown in Fig. 6. The bulk of the remainder is redeposited as film on the piston (Figs. 3 and 8).

OPEN-VALVE INJECTION RESULTS – We now examine the results of the OVI calculation and compare them with the CVI case. The OVI case required only one engine cycle to visualize the trend, in contrast to the CVI case, which required the calculation of two engine cycles to follow the full intake process.

Figure 9 shows the computed film on the bottom of the head and valves at four different crank angles in the engine cycle for the OVI case. A representative experimental image is shown for comparison in Fig. 10. In contrast to the CVI case, it is evident in both figures that films are formed on, between, and downstream of the exhaust valves. Figure 9 indicates that the films are thickest at 360 CAD (TDC compression). The experimental observations concerning iso-octane film location on the head for the CVI and OVI cases are also in general agreement with those observed when California Phase 2 gasoline is used for the fuel (Fig. 11).

The conclusions from Figs. 9 and 10 are substantiated and quantified by graphs from the KIVA-3V OVI calculations. Figure 12 shows the computed history of fuel in the cylinder for the OVI case, and Fig. 13 is the film deposition history. We observe that although over half the fuel charge is deposited on the tops and stems of the intake valves, a significant amount of spray nevertheless succeeds in entering the cylinder. However, much of it is deposited as film, with 2 mg in localized thick regions on the downstream edges of the intake valve seats. Nearly 3.5 mg are deposited on the exhaust valve faces with lesser amounts on the head and on the cylinder liner beyond the exhaust valves.

Note in Fig. 12 that less than 0.5 mg of fuel vapor is formed in the cylinder after intake valve closure (IVC), and that there is only 1.7 mg of fuel vapor at an approximate ignition time (330 CAD). This is in contrast to 3.5 mg at 1050 CAD in the CVI calculation (Fig. 6). Correspondingly, the actual average equivalence ratio in the cylinder is halved from the 0.15 of the CVI case to

0.075 in the OVI case. As in the CVI case, all but about 0.5 mg of the fuel vapor in the cylinder is lost during the exhaust stroke. The total fuel (liquid and vapor) in the cylinder in the OVI case is nearly 10 mg, but most of this is in the form of films.

Figures 12 and 13 also illustrate the transfer of wall film from the liner to the piston as the piston approaches TDC, and the reversal of this process as the piston goes back down. The presence of wall wetting is an undesirable feature in the OVI case. For this particular geometry and OVI fueling schedule, it appears that the effects of wall splash are not significantly increasing the in-cylinder fuel vapor concentrations. The greater film vaporization in the CVI case implies that the engine would start sooner were it being fired.

SUMMARY

Recent extensions of the wall film model of the KIVA-3V computer program are reported. The new wall film model retains calculation methods that were reported in a previous SAE paper [1] for film transport on complex surfaces, heating and vaporization of the film, and separation and re-entrainment of film at sharp corners. The extensions allow prediction of the effects of spray/wall interaction, including droplet splash and impingement-pressure spreading. The extensions also include a submodel for expelling liquid from valve seat areas when valves close.

The results of example calculations of open- and closed-valve injection are then reported. Computed liquid film locations agree qualitatively with locations experimentally measured using an LIF technique [3]. The calculations also show that, although more fuel can enter the cylinder with open-valve injection, under cold-start conditions most of the fuel in the cylinder may reside in wall films, and consequently there may be less fuel vapor in the cylinder than with closed-valve injection.

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ENGINE TIMINGS

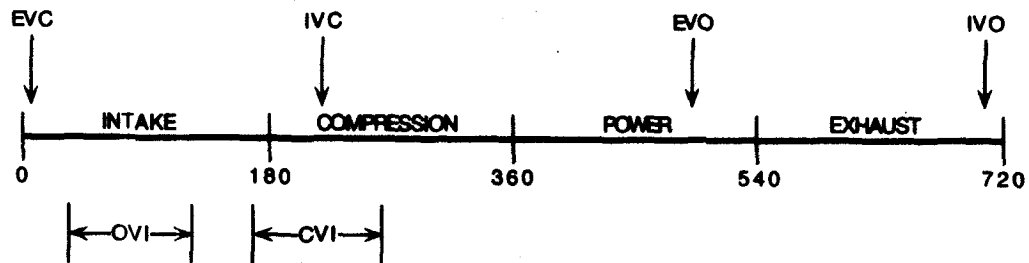


Figure 1. Valve and injection timings for the DOHC engine.

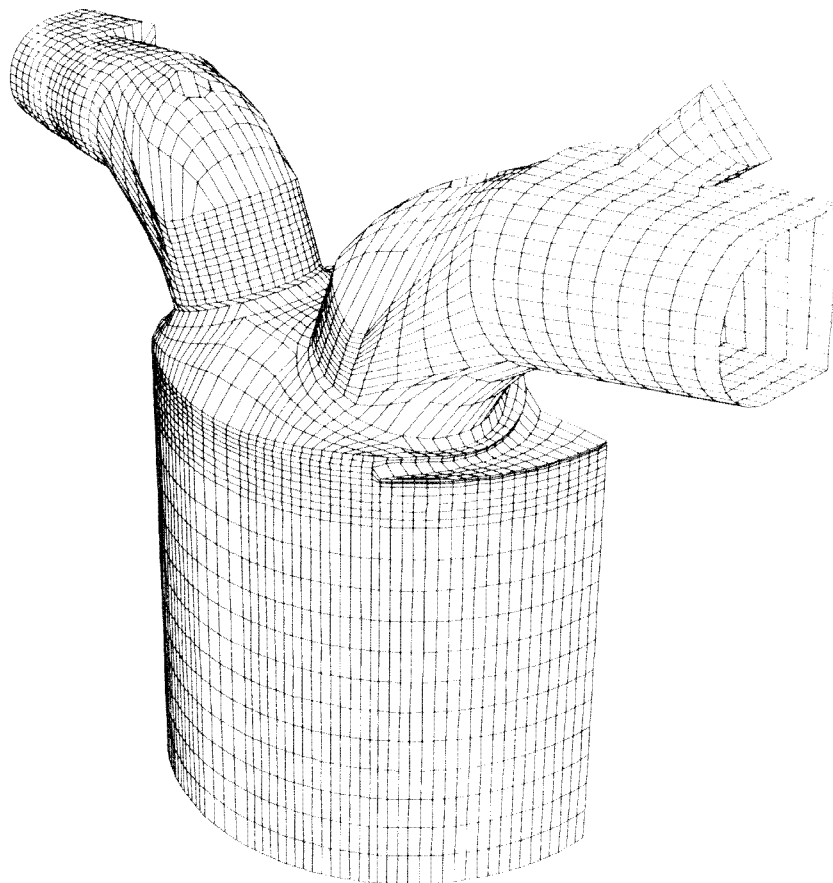


Figure 2. Perspective view of the KIVA-3V computational mesh for the 4-valve DOHC engine used in the PFI study. Only one half the geometry is zoned, to take advantage of symmetry.

DUAL SPRAY CVI

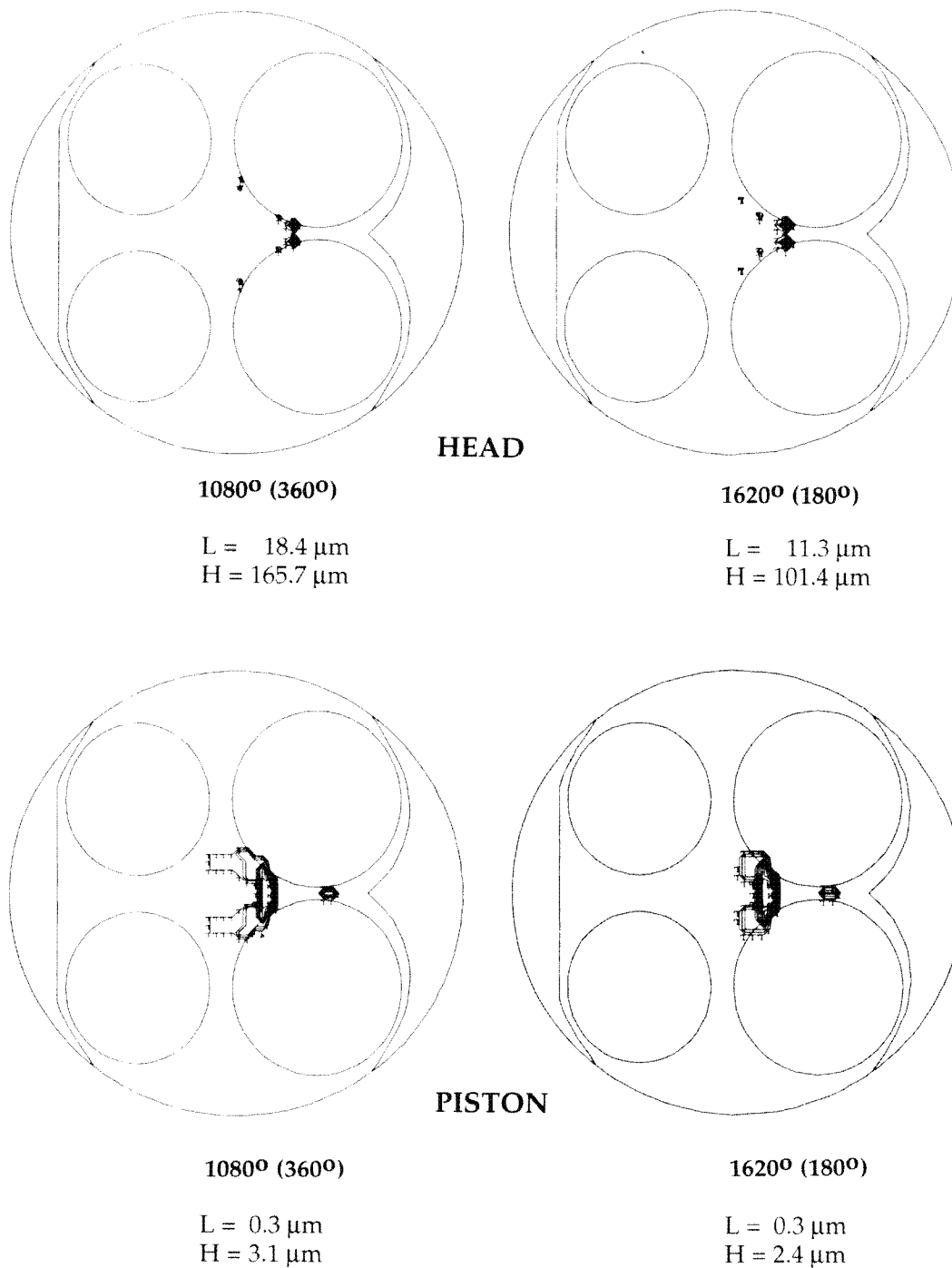


Figure 3. CVI: computed film deposition on the head and piston surfaces at two different crank angles in the engine cycle.

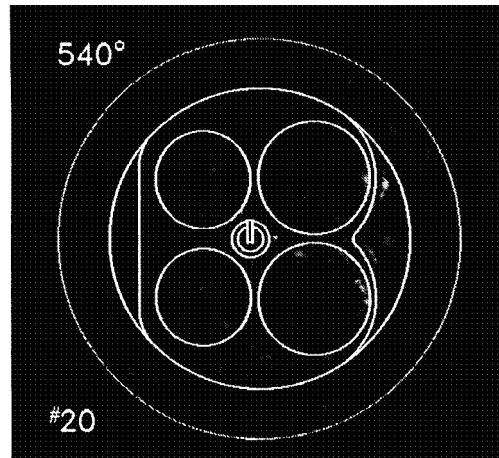


Figure 4. CVI: experimental LIF image of film on the head and piston surfaces at BDC intake.

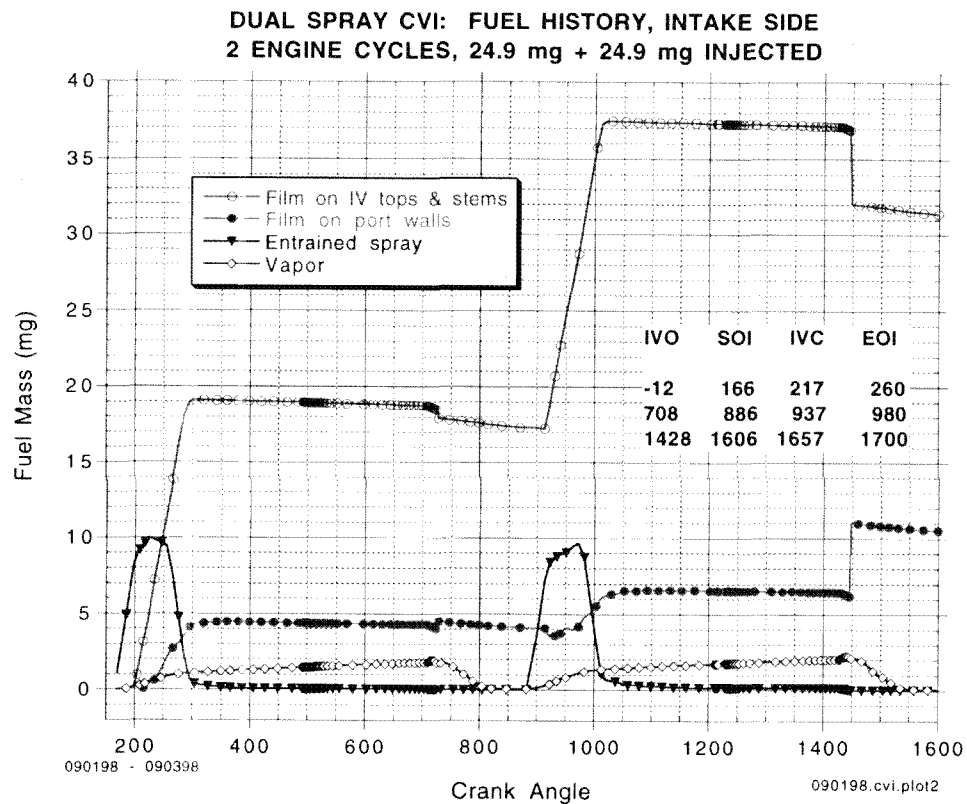


Figure 5. CVI: computed fuel history on the intake side.

DUAL SPRAY CVI: ISOCTANE VAPOR HISTORY, 2 ENGINE CYCLES

SOI = 166 CAD, EOI = 260 CAD, 24.9 mg INJECTED,

SOI = 886 CAD, EOI = 980 CAD, 24.9 mg INJECTED

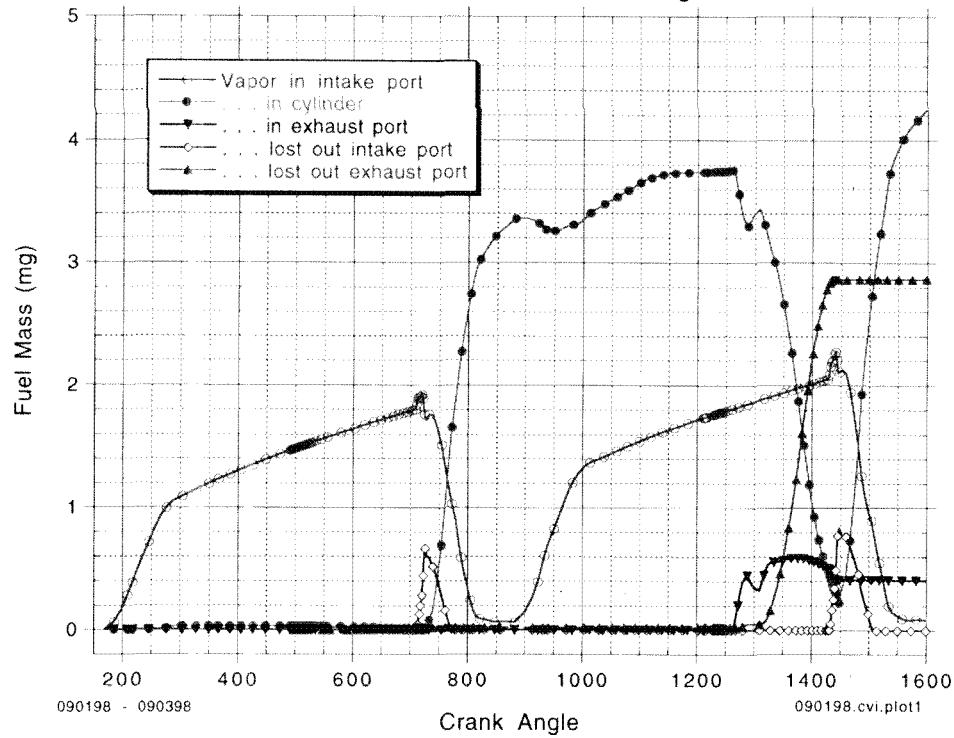


Figure 6. CVI: computed vapor history.

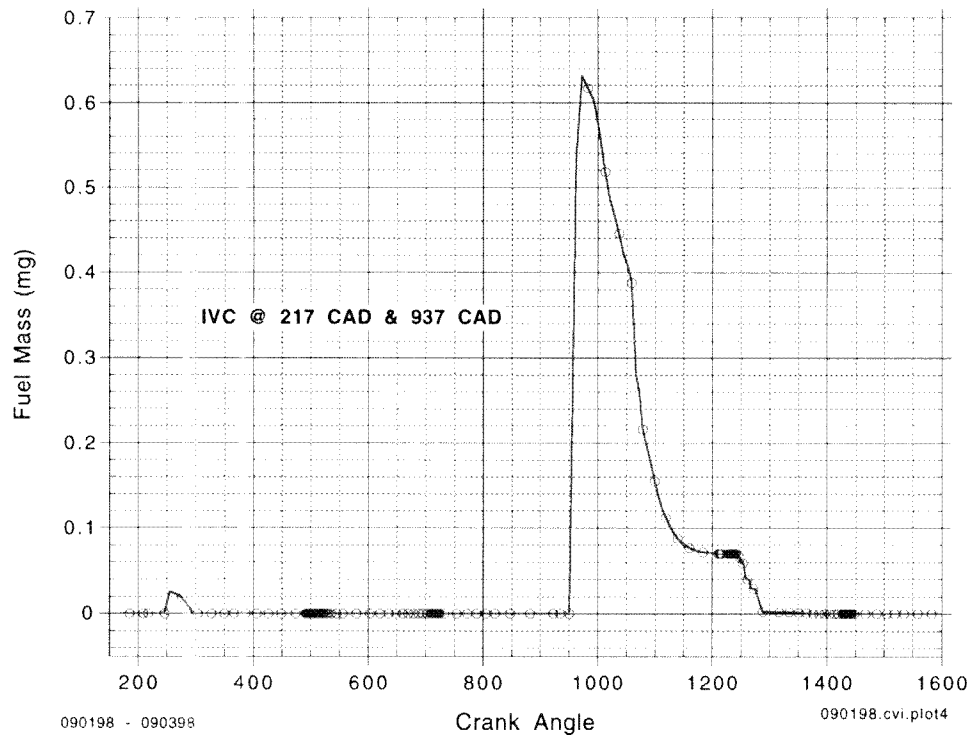
DUAL SPRAY CVI: SPRAY ENTRAINMENT IN CYLINDER

Figure 7. CVI: computed spray entrained in the cylinder.

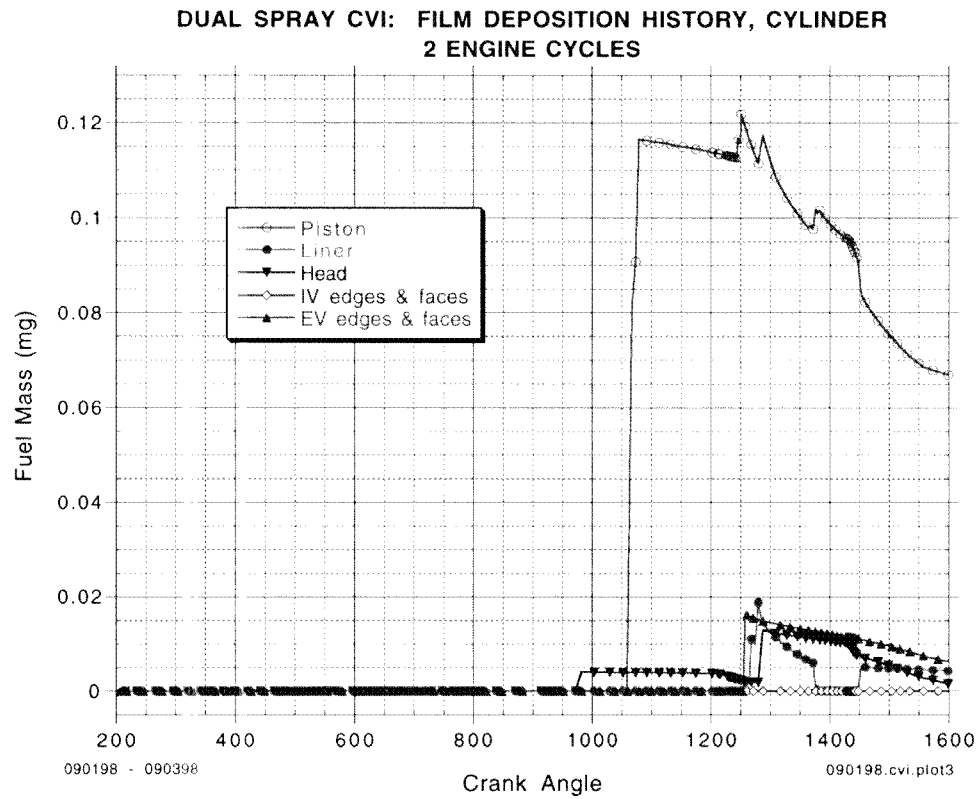


Figure 8. CVI: computed film deposition history in the cylinder.

DUAL SPRAY OVI

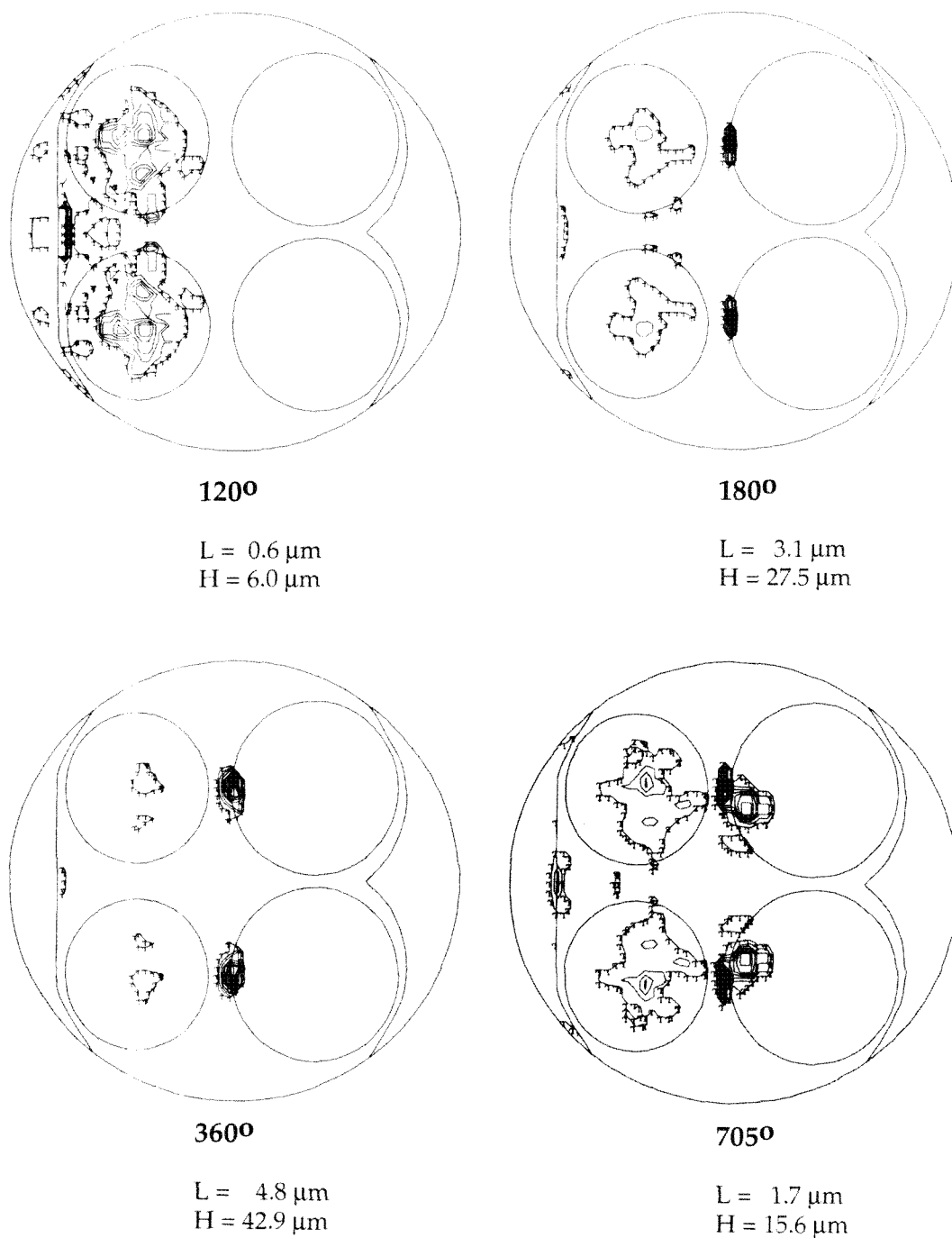


Figure 9. OVI: computed film deposition on the head surfaces at four different crank angles in the engine cycle.

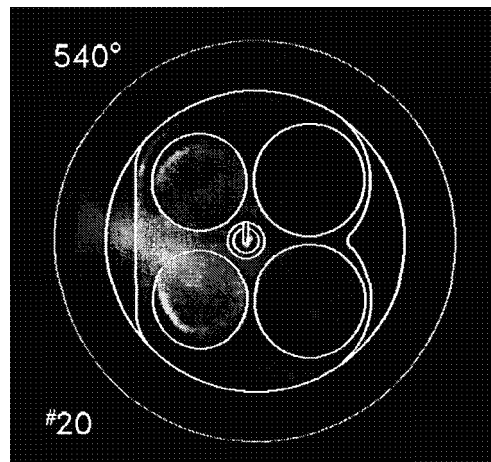


Figure 10. OVI: experimental LIF image of film on the head at BDC intake.

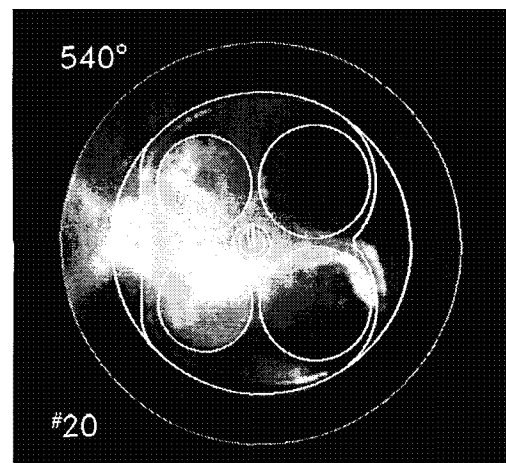
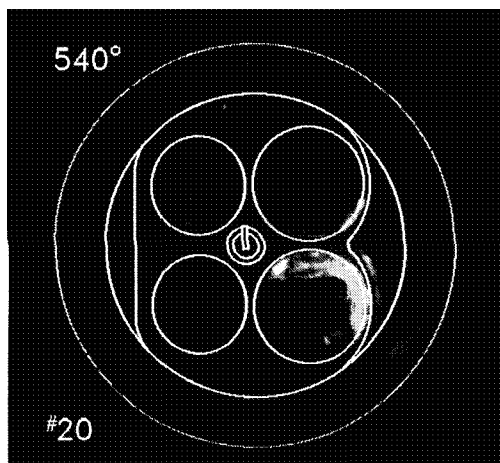


Figure 11. CVI (left) and OVI (right): experimental LIF images for California Phase 2 gasoline, at BDC intake.

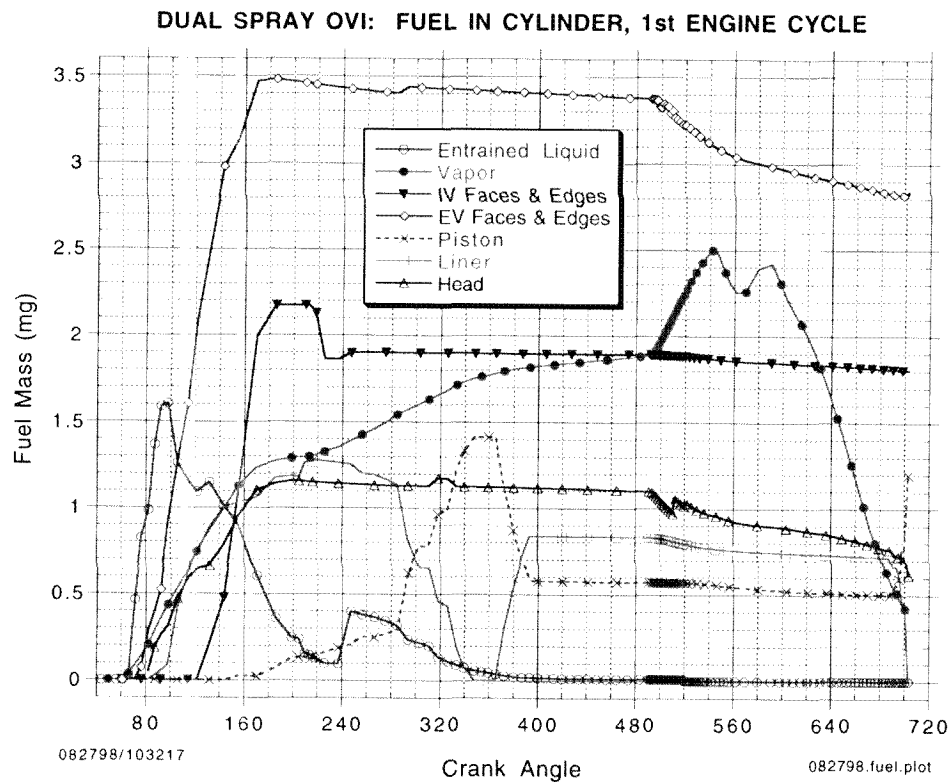


Figure 12. OVI: computed fuel history in the cylinder.

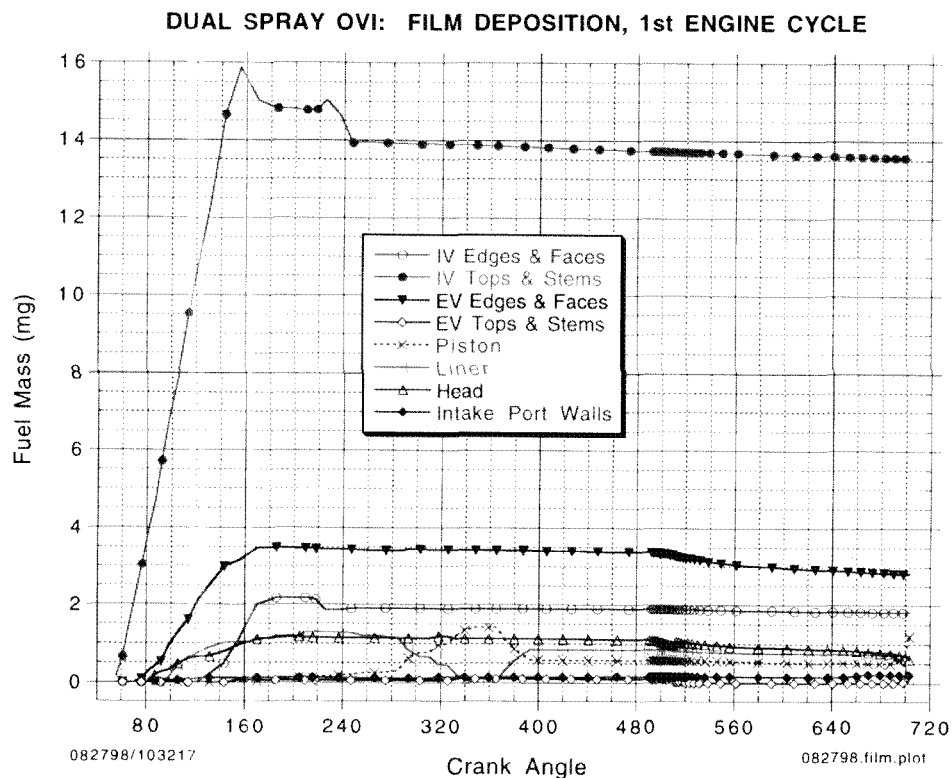


Figure 13. OVI: computed fuel deposition history.